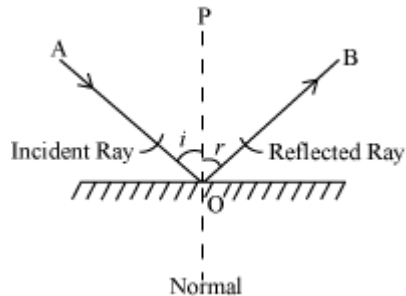


Light

- Reflection of light makes things visible.

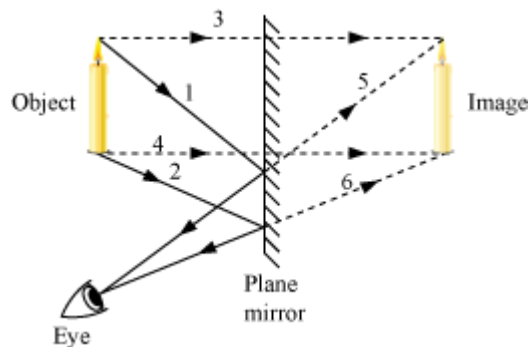


- (a) i (Angle of incidence) = r (Angle of reflection)
- (b) AO, OP, and OB lie on the same plane.

- **Laws of reflection:**

- The angle of incidence is equal to the angle of reflection
- The incident ray, the normal at the point of incidence and the reflected ray all lie in the same plane.

- **Image formation by a plane mirror**



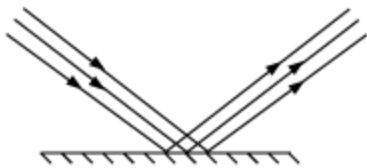
Left part of the candle appears on the right and its right part appears on the left. This is known as **lateral inversion**.

- **Characteristics of images formed by plane mirror**

- virtual and erect
- same size as of object
- laterally inverted
- image distance and object distance are same and perpendicular from mirror

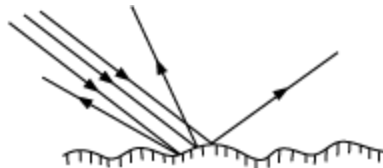
- Virtual images are those images which cannot be obtained on screen. But there are some images which can be obtained on screen. Such images are called real image.
- **Uses of plane mirror**
 - It is used as a looking glass.
 - It is used to increase the effective length of an optician's room.
 - In periscope, two parallel plane mirrors are inclined at 45 degrees with vertical walls such that they are facing each other.
 - In kaleidoscope, three plane mirrors are inclined with each other at 60 degrees.
 - It is used in solar heaters and cookers to heat substances by reflecting the sunlight towards the substances.

○ **Regular reflection**



Regular reflection

Irregular and diffused reflection



Diffused reflection

- The laws of reflection are valid in regular as well as irregular or diffused reflections.
- Smooth or polished surfaces gives regular reflection.
- Uneven or unpolished surfaces gives irregular reflection.
- Objects that give their own light are known as **luminous objects**
- Objects that are visible because of reflected light are known as **illuminated objects**.
- Kaleidoscope works on the principle of **multiple reflections**.
- **Periscope** is an optical device used to see objects that are not along the line of sight.
- Sunlight consists of several colours.
- Splitting of white light into different colours is called dispersion.

- Number of image formed in multiple reflection = $\frac{360^\circ}{\text{angle between two plane mirrors}} - 1$

- Vitamin **A** (raw carrots, broccoli, green vegetables, cod-liver oil, etc.) is necessary for good vision.
- **To protect your eyes, the following points should be remembered:**
 - Avoid reading in dim light.
 - Wash your eyes at least four times a day with clean and cold water.
 - Wash your eyes quickly if dust particles or small insects enter your eye.
 - Visit an eye specialist regularly. Improper vision can cause stress, eyestrain, and even headaches.
 - While reading, maintain a distance of atleast 25 cm between your eyes and the book.
 - Do not rub your eyes. If redness in the eye persists, then consult an eye specialist immediately.
 - Avoid direct exposure to sunlight. Exposure to a large amount of light can harm your retina.
- Some persons cannot see since birth and some may loose their eye sight because of a disease.
- Visually challenged persons try to identify things by touching and listening.
- **Braille system** is helpful for visually challenged persons.
- Many Indian languages can be read using the Braille system.